Exploring biomarker dynamics in *inflamm-aging* phenomenon: A longitudinal multivariate analysis from the HEBE project

UNIVERSITÀ DEGLI STUDI DI MILANO

HEBE SALUTE IN MOVIMENTO

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INTRODUCTION

Aging is closely linked to an increased risk of non-communicable diseases[1] (NCDs), driven in part by ***inflamm-aging***[2], a state of chronic low-grade inflammation.

Exercise / physical activity is widely recognized as a modifiable factor capable of mitigating the effects of *inflamm-aging*. However, understanding the complex interactions among the biomarkers involved requires integrative approaches that go beyond traditional analyses.

The aims of the **HEBE project**[2] include:

1. Investigating the influence of tailored exercise interventions on the dynamics of key biomarkers
2. Offering insights into strategies for healthier aging and the prevention of age-related decline.

OBJECTIVES OF THE CURRENT STUDY

The **primary objective** of this study was to evaluate the impact of a personalized physical exercise program on biomarkers associated with *inflamm-aging*, e.g. anti/inflammatory cytokines. By assessing changes before and after the intervention, the study aimed to understand how targeted lifestyle modifications can reduce chronic low-grade inflammation, a key factor in cellular and systemic dysfunction.

A **secondary objective** was to examine the downstream effects of these changes on aging-related outcomes, such as telomere length and oxidative stress levels, to explore how reductions in inflammation may influence cellular aging and oxidative balance.

Finally, the study sought to integrate these findings into a comprehensive statistical framework, offering evidence-based insights into how exercise and lifestyle modifications can mitigate the adverse effects of *inflamm-aging* and support healthier aging trajectories.

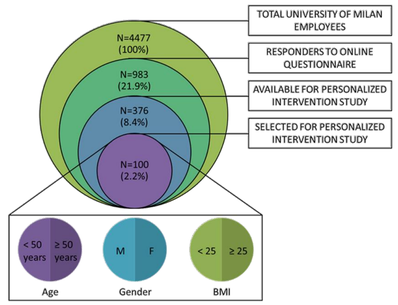
MATERIALS & METHODS

One hundred participants, recruited via email among employees of the University of Milan, were selected based on a questionnaire assessing their quality of life and psychophysical well-being. These participants agreed to take part in a six-month intervention study.

Each participant followed a personalized exercise program. Blood samples were collected at baseline (T0) and after the intervention (T1) to measure biomarkers, including cytokines, telomere length, and oxidative stress markers, and assess their relation with clinical data.

Principal Component Analysis (**PCA**) was used to explore global patterns among biomarkers at both time-points, while Repeated Measures ANOVA-Simultaneous Component Analysis (**RM-ASCA**) was applied to model temporal trends.

The study was approved by the ethics committee of UNIMI, and all participants provided informed consent.



CONCLUSIONS

The observed alterations in biomarker levels following a 6-month physical activity intervention underscore the body's multifaceted physiological adaptations to tailored exercise. Elevations in markers such as haptoglobin, MCP-2, BDNF, GDF-15, IL-8, IFN-γ, and PAI-1 suggest **enhanced acute-phase responses, neuroplasticity, and immune system activation**. Conversely, reductions in FGF-2, FGF-23, MCP-1, irisin, IL-18, TNF-a, IL-10, VEGF-A, IL-6, IL-1ra, leptin, and insulin suggest **diminished chronic inflammation, improved metabolic regulation, and refined vascular function**. Collectively, these biomarker shifts reflect the complex interplay between inflammatory, immune, metabolic, and neurotrophic processes, highlighting the systemic benefits of regular physical activity.

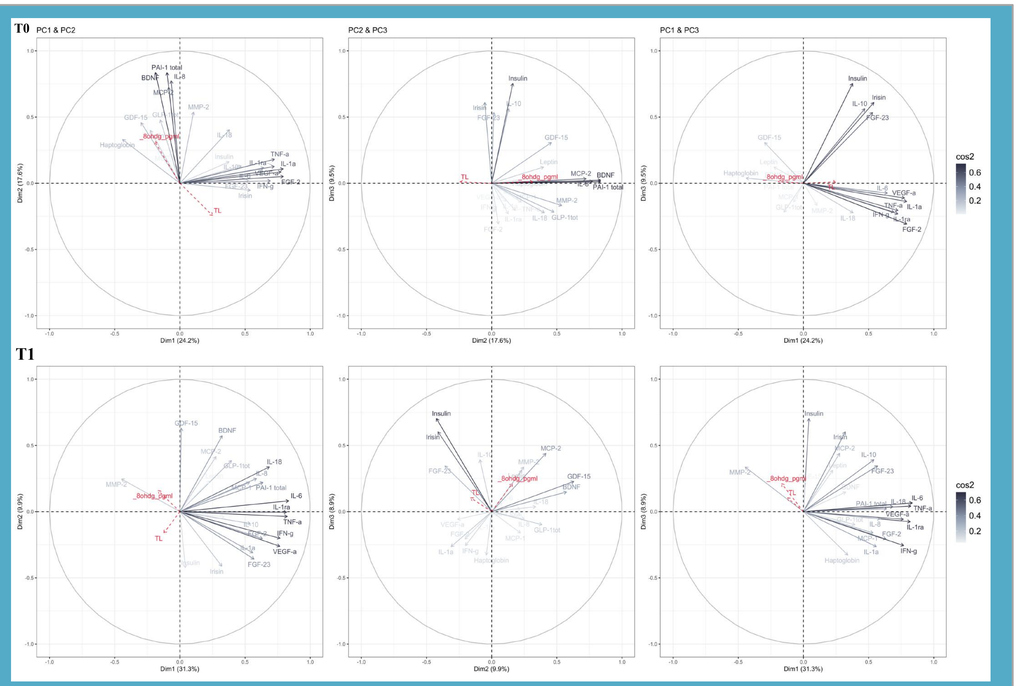
Limitations

Further investigations are required to identify the best imputation strategy for dealing with values out-of-range for quantitation. A more robust and accurate method would involve employing censored distribution techniques, specifically designed to manage data with detection limits.

PCA RESULTS

At T0 (first row), PC1 captured variability primarily among a core group of inflammatory markers, including **TNF-a**, **IL-1(r)a**, **VEGF-a**, **FGF-2**, emphasizing their central role in systemic inflammation. PC2 highlighted a distinct cluster of markers, such as **PAI-1**, **BDNF**, **MCP-2**, and **IL-8**, reflecting indipendent potential association between neurotrophic and inflammatory pathways, immune cell recruitment, and senescence. PC3 emerges with an indipendent group composed by insulin, irisin, IL-10 and FGF-23.

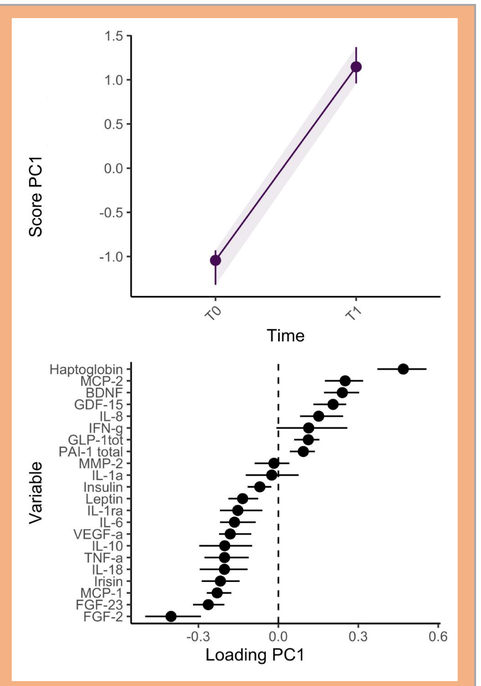
After physical exercise intervention (T1), PC1 remained predominantly associated with the same inflammatory cytokines. However, a weaker clustering of these markers was observed, suggesting a less cohesive inflammatory profile. Additionally, the distinct separation between markers on PC1 and PC2 seen at T0 disappeared.



RM-ASCA RESULTS

PC1 extracted two general patterns of change associated with positive and negative loadings: this represents an increase (for positive loadings) or decrease (for negative loadings).

Higher levels of haptoglobin, **MCP-2**, **BDNF**, **GDF-15**, **IL-8**, **IFN-γ**, and **PAI-1** emerged while decreased levels of **FGF-2**, **FGF-23**, **MCP-1**, **irisin**, **IL-18**, **TNF-a**, **IL-10**, **VEGF-A**, **IL-6**, **IL-1ra**, **leptin**, and **insulin** emerged. These biomarker changes highlight the complex physiological responses to physical activity, encompassing inflammatory, immune, metabolic, and vascular adaptations.



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2. Bianchi, F. *et al.* HEBE project: Healthy aging versus inflamm-aging: The role of physical exercise in modulating the biomarkers of age-associated and environmentally determined chronic diseases, study protocol. *PLoS One* 19, (2024)

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